جامعة بيرزيت

BIRZEIT UNIVERSITY

FACULTY OF ENGINEERING AND TECHNOLOGY

ELECTRICAL AND COMPUTER ENGINEERING DEPARTMENT

CHIP DESIGN VERIFICATION – ENCS5337

Course project - Birzeit Integrated Router Design (BIRD)

Objective:

This is the ENCS5337 course project. The project is about building a complete verification environment for **BIRD: Birzeit Integrated Router Design**, a packet-based routing block.

The attached project specification is your main reference for the design verification requirements. You can also find the DUT and a simple test demonstrating basic functionality at:

https://www.edaplayground.com/x/UtWv

You are required to provide:

1. A test plan. Excel format is acceptable.
2. For each item in the test plan, clearly list the test name that will be implemented. A single test may cover one or more test plan items.
3. A full testbench including (you are not restricted):
    a. Interface and clocking blocks
    b. Environment
    c. Agents
    d. Checkers
    e. Drivers
    f. Monitors
    g. Sequences
    h. End-of-test reporting
    i. Tests
4. Code coverage and functional coverage reports.

Deliverables:

All code must be uploaded to a GitHub repository, and a copy must also be available as a Git client on your EDA server account.

The main project Git client folder must be named as follows:

BIRD_student_id   For example: **BIRD_1220167**

You must follow the directory structure used in the example discussed in class. If you suggest a different structure, you must be able to defend your decision during the discussion.

The following must be included in the project folder:

- All testbench code
- Test plan file
- Code coverage report
- Functional coverage report

Student collaboration will be evaluated through GitHub commits. Therefore, I should be able to clearly see which part of the code each student contributed.

Groups:

Each group must consist of **4 students**. Smaller groups are not allowed unless the total number of students in the class is not divided evenly. Group members may be mixed from the two sections.

Grading:

Evaluation will be based on the test plan, code, and coverage reports. However, there will also be a discussion session for each group. Based on the discussion and your understanding of your deliverables, the grade will be assigned on an individual basis.

Deadline:

- The deadline is: 20/6/2026 at 23:59
- Any change to the server folder or GitHub repository after the deadline will be considered a late submission.
- Late submissions will be penalized at a rate of 10% of the project mark per day.

Plagiarism:

Any type of plagiarism or cheating will result in a mark of zero. Cheating cases will also be handled according to university regulations.